

Linux Interview Questions

1. what is Linux, and how it is different from UNIX?

=====

Linux and Unix are both strong operating systems, but they are designed for different purposes and audience

Linux:--> linux is free open source and widely used on everything from personal computers to servers, and even in smart phones

--> it is constantly improved by a large community of developers which means faster updates and bug fixes

UNIX:

--> unix on the other hand is a proprietary system mainly used in enterprise environment like large servers and workstation because it is not

--> not open source system

--> we need a license to use it

--> and updated tends to be slower

flexibility:

--> linux supports a wide range of the file systems and hardware, making it super versatile

--> unix is more specialized and tailored for specific enterprise-level tasks

linux

open source and accessible to everyone

perfect for personal and professional use

UNIX

unix is more exclusive used in large organizations

for stability and dedicated support

2. what is linux kernel, why it is important?

=====

--> The linux kernel is the brain of the linux operating system. Its the core part that connect your hardware and software managing key sources like CPU, memory and devices to keep everything running smoothly

-->it handles important tasks like running multiple programs at once like process management

--> organizing files and storage

--> managing network communication

--> one thing about the kernel is the hardware Abstraction
this means software doesn't need to know the details of your hardware, its just communicates with the kernel, this makes Linux both flexible and efficient

--> another great feature is Modularity
you can add extra features like device driver or network protocol without messing with the main system.

--> this makes the linux super customizable whether its for personal use or powering servers

--> inshort kernel is essential for making sure that your hardware and software work together seamlessly

3. what is a shell in linux, and how it is different from bash?

=====

Shell

--> shell is basically a command-line interface that lets you interact with the operating system

--> you can think of it is a bridge between you and systems kernel. it takes the command you type, interprets them and executes them

--> shell comes handy with features like

scripting

command history

job control

--> it helps you automate the task and manage process more efficiently

--> power shells in linux is

Bash

Zsh

Fish

Bash (Bourne Again Shell)

--> Bash is one of the most commonly used shell.

--> it is the improved version of original bourne shell (sh) and comes with extra features like

better scripting tools

command line editing

support for arrays and functions

shell is general term for any command line interface

bash is a specific type of shell with advances features

if your'e looking for something powerful and user friendly for scripting or command line task, bash is a great choice

4. what are the basic components of a linux OS?

=====

linux operating system is built on five key components each essential for smooth operation

1. kernel

--> the core of linux, meaning it manages hardware resources and rich hardware and software.

--> it handles like memory management, process scheduling, and device communication

2. system library

--> these library provide function that allow application to process kernel feature without directly interacting with it

--> so the symplify software development by implementing core functionality making the development more efficinet

3. system utility

--> specialized program handle tasks like

file management

system monitoring

configuration supporting daily operation

4. Bootloader

--> it initializes the system by loading the kernel into memory at startup, enabling boot up, meaning its essential for enabling the system to boot up

5. application software

--> program like browsers or editors installed by user often via app stores and linux distribution like ubuntu

5. what is the init process in linux?

=====

--> the init process in linux is like the systems manager. it takes over as soon as your computer starts

--> when you turn on your linux machine, the kernel or the core of the OS gets loaded into the memory and starts working

--> but the kernel doesn't handle users process or services, that's where init comes in

--> and it is the very first process the kernel starts and it always has a process ID of 1

--> its main job is to set the system and manage everything that runs after booting.

--> this includes first starting essential services

network settings

background tasks

--> setting the systems mode

using a graphical interface

or command line

--> keeping an eye on process and restarting them if they crash

--> init has been replaced by systemd
which is faster, more efficient and offers greater flexibility

6. how do you find files in linux?

=====

to find files in linux - find command is used

find [path] [options] [expression]

path: starting point (eg: /home/user)

options: file type (-type f) or directory (-type d)

expression: conditions like -name "*.text", or -size +1M

firstly to find the file, we have to create the file

--> for creating the file use touch command

--> touch file1.txt

--> to find the directory use pwd command

--> pwd

--> to see where this file has been created

--> find /c/users/2142289 -type f -name "file1.txt"

```
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```

```
$ touch file1.txt
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ pwd
```

```
/c/Users/2142289
```

```
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```

```
$ find /c/Users/2142289/ -type f -name "file1.txt"
```

```
/c/Users/2142289/file1.txt
```

7. what is the difference between soft link and hard link?

=====

Hard link and soft link are the two ways to create references to file in linux or unix systems

Hard link:

--> its like having two identical names for the same file, both named directly point to the actual data on your storage

--> hard links only work within the same storage area

for example file system and you cant use them to link folders unless you are system administrator

soft link:

--> a soft link or a symbolic link is more like a shortcut you create on your desktop. it points to the files location rather than the actual data

--> for example if the original file is moved or deleted. the soft link stop working because it does not contain the actual data and it just contain the files address

Hard link:

--> a direct reference to the files data on disk

--> must be on the same file system

--> file remains accessible until all the hard link are deleted

--> in target link_name

--> share the same inode with the original file

soft link:

--> a pointer to the original file

--> can link across different file system--> deleting the original file breaks the link

--> ln -s target link_name

--> require additional inode for the link

8. how do you change file permissions in linux using the CHMOD command?

=====

to change the file permissions in linux to use the CHMOD command

for read --> (r) --> owner

for write --> (w) --> group

for execute --> (x) --> other

symbolic mode

--> chmod u+x file.txt

adds excute permission for the owner (u)

--> chmod g-w file.txt

removes write permission for the group (g)

--> chmod o=r file.txt

sets read-only permission for others (o)

numeric mode

--> chmod 755 file.txt

owner: read/write/execute, group & others: read/execute

--> if u want to give the read permission to a file will write

chmod o=r file1.txt

--> similarly for the execution you can give

chmod u+x file1.txt

--> for the write permissions

chmod g-w file.txt

9. what are the different types of permissions available for files in linux?

=====

linux file permissions

who can access?

details:

owner - creator of the file

group - a specific group of users

others - everyone else

what can they do?

details:

read (r) - view file content

write (w) - modify the content

execute (x) - run the file as a program

r - 4

w - 2

x - 1

how are permissions shown

details:

rw- r-- r--

first character: file type (- for file, d for directory)

next 9 characters: permission for owner, group and others)

10. how do you create and manage symbolic links?

=====

so symbolic links also known as symlinks or softlinks are the files in linux that point to other files or directory we just read what are soft links and hard links and that symbolic link. the same soft link is known as symbolic link

--> so to create the symbolic link will use

```
ln -s /c/Users/2142289/file1.txt /c/Users/2142289/first1_link.txt
```

if you want write and save the changes in first1_link.txt, the changes will displayed in file1.txt
discard and reload the changes you will see the changes in file1.txt

create a symbolic links

cmd:

```
ln -s [target] [link_name]
```

explanation:

creates a shortcut to the target file or directory

example

cmd:

```
ln -s /var/www/html my_website
```

explanation:

links my_website to /var/www/html

view symbolic links

cmd:

```
ls -l
```

explanation:

displays symbolic link with ->

remove a symbolic link

cmd:

```
rm [link_name]
```

explanation:

deletes the link without affecting the target

update a symbolic link

cmd:

```
ln -sf [new_target][link_name]
```

explanation:

updates the link to point to a new target

11. how do you check your current path/directory?

=====

command : pwd

purpose : displays the absolute path of your current working directory

example : \$pwd

/home/username/Documents

12. how do you combine two commands, and what is the use of a pipe (|) in linux?

=====

--> in linux you can combine two commands and use a pipe to send the output of one command as an output of another

--> what is the pipe (|) operator?

--> the pipe (|) is used to send the output of one command directly as input to another command

--> this allows you to chain commands together, enabling more complex tasks in a single line

command 1 | command 2

suppose you want to list all the files in the directory and search for a specific file name

command:

ls | grep file1.txt
ls

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\$ ls | grep file1.txt

file1.txt

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\$ ls

13. how can you check for free disk space?

=====

df command - which stands for (disk free) and shows available and use disk space on the system

df -h --> shows disk space in human readable format

check disk space usage

df - shows total space, used space, available space, and usage percentage

human-readable format

df -h --> space is shown in GB or MB for easier understanding

check directory space usage

du -sh [directory] --> shows the disk usage of a specific directory in a readable format

14. write a command to find files with the .txt extension containing a specific string?

=====

syntax:

grep -rl "search_string" --include="*.txt" /path/to/search

example:

grep -rl "error" --include="*.txt" /home/user/documents

this command will search for the string "error" in all .txt files in the /home/user/documents directory and its subdirectories, and will output the filenames of those that contain the string

grep -lr "hey" --include="*.txt" /c/Users/2142289/

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\$ grep -lr "hey" --include="*.txt" /c/Users/2142289/

15. what are the different ways to view the content of a file without using the cat command?

=====

less --> less filename --> view file content page by page

more --> more filename --> view file content, forward only

head --> head filename --> view the first 10 lines of a file

tail --> tail filename --> view the last 10 lines of a file

nl --> nl filename --> view content with line numbers

16. how do you check the current IP address of your linux server?

=====

command : ip a

--> (ip a) is a part of the ip suite commands used to display network interfaces and their configurations

--> it provides detailed information about all network interfaces on your linux system, including their IP addresses, MAC addresses, interface status, and more

```
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```

```
$ ip a
```

17. what is SSH, and how it is used to access a linux server remotely?

=====

--> SSH (Secure Shell) is a cryptographic protocol used for secure remote access and command execution on linux servers

--> it encrypts communication to prevent eavesdropping and attacks

the basic syntax for using the SSH command is as follows:

```
ssh [username]@[hostname or IP address]
```

here, replace [username] with your remote server username, [hostname or IP address] with the servers hostname or IP address

18. what is package manager in Linux, and why it is useful?

=====

--> a package manager in linux is a tool that automates the process of installing, updating, configuring, and removing software packages (applications or libraries)

--> it simplifies software management by handling dependencies, version control, and updates

--> some of the widely used package managers in linux

1. apt (advanced package tool)
2. yum (yellow dog updater)
3. dnf (Dandified YUM)

19. how do you terminate an ongoing process in linux?

=====

kill

--> the kill command sends a signal to a process, instructing it to terminate

basic syntax:

kill [options] <PID>

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\$ kill

kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]

20. how do you check system architecture and CPU/memory stats?

=====

--> A command-line utility "lscpu" in linux is used to get CPU information of the system. The "lscpu" command fetches the CPU architecture information from the "sysfs" and /proc/cpuinfo files and displays it in terminal

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\$ vmstat -d

1. what is the use of echo command?

=====

--> echo is a command that outputs the strings that are passed to it as arguments

echo "Hello World"

```
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$ echo "Hello World"  
Hello World
```

2. how to check the computer name or hostname in linux

=====

--> using 'hostname' command

```
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$ hostname  
LTIN531281
```

3. how to check the current user in linux terminal?

=====

--> whoami

```
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$ whoami  
CTS+2142289
```

4. how to check your current path/directory you are working in?

=====

--> pwd (present working directory)

```
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$ pwd  
/c/Users/2142289
```

5. explain the difference between relative and absolute path?

=====

--> relative path : start from current working directory

```
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$ cd My\ Documents/
```

--> absolute path : the full path to a file or directory

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\$ cd /c/Users/2142289/My\ Documents/

6. which command to be used to create a file in linux?

=====

1. touch

2. vi

3. vim

4. nano...

--> touch myfile

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\$ touch myfile

\$ ls

7. how will you edit an existing file on a linux server?

=====

using vi, vim, nano etc

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\$ vi myfile

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\$ cat myfile

Hi, This is sample file

8. how to rename a file in linux?

=====

using the (mv) command

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\$ mv myfile new_file

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\$ ls

9. how to search for a string in a file?

=====

using grep command

```
grep "your_string" filename.txt
```

```
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$ grep "hi" new_file  
Hi, This is sample file
```

```
CTS+2142289@LTIN531281 MINGW64 ~  
$ grep "sample" new_file  
Hi, This is sample file
```

10. difference between grep and egrep?

=====

In egrep, you can search for more than one strings at same time

```
egrep "key1|key2|key3"
```

```
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$ egrep "hi|sample" new_file  
Hi, This is sample file
```

11. how can you read a file without using cat command?

=====

using less, more, vi, command

less filename

```
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$ less new_file
```

12. what is the advantage of using less command?

=====

- > we can easily read big files
- > forward and backward search is easy
- > navigation from top to bottom is easy

13. how to check a file's permission?

=====

```
ls -l  
ll  
getfacl file_name
```

14. how to check the IP of you linux server?

=====

ifconfig

ip addr

15. how to read the top 5 files in a file?

=====

using head command

head -5 file_name

16. how to read the last 5 files in a file?

=====

using tail command

tail -5 file_name

17. how to list hidden files?

=====

ls -a

ls -la

18. how to see all the recently used commands?

=====

history command

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\$ history

19. what is root?

=====

--> root is admin or super user

--> /root home directory for root user

--> /root directory

20. what is inode and how to find it for a file?

=====

ls -li

inode is index node. it server as a unique identfier for a specific piece of metadata on a give filesystem

21. which command can you use for finding files on a linux system

=====

using find and locate command

find current_directory -name file_name

find . -name new_file

22. command for counting words and lines

=====

wc (word count)

wc -l (word count line)

wc filename

wc -l filename

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\$ wc -l new_file

9 new_file

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\$ wc new_file

9 13 52 new_file

23. how can you combine two commands or what is pipe command used for?

=====

we can combine two commands using pipe (|)

ex: command1 | command2

pipe is used for to combine two commands and redirect output of command1 to command2

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\$ cat new_file | wc -l

9

24. how to view the difference between two files?

=====

diff file1 file2

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\$ diff file1.txt new_file

0a1,9

> Hi, This is sample file

> Hi

> hey

> hi

> hey

> hi

> hey

> hi

> hey

25. what si the use of shred command do? (permanently delete a file which is unable to recover)

=====

permanently delete a file which is unable to recover

shred -u file_name

shred --remove file_name

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\$ shred -u first1_link.txt

26. how to check system architecture info?

=====

dmidecode and lscpu

27. how to combine two files?

=====

the content of file1 and file2 will combine and write it in file3

cat file1 file2

cat file1 file2 > file3

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```
$ cat file1.txt new_file > newfile3
```

```
$ cat newfile3
```

```
Hi this is sample text
```

```
Hi, This is sample file
```

```
Hi
```

```
hey
```

```
hi
```

```
hey
```

```
hi
```

```
hey
```

```
hi
```

```
hey
```

28. how can you find the type of file?

=====

```
file file_name
```

```
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```

```
$ file newfile3
```

```
newfile3: ASCII text
```

```
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```

```
$ file AppData
```

```
AppData: directory
```

29. how to sort the content of a file?

=====

```
sort file_name
```

```
cat file_name | sort
```

```
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```

```
$ cat newfile3 | sort
```

```
Hi
```

```
Hi this is sample text
```

```
Hi, This is sample file
```

```
hey
```

```
hey
```

```
hey
```

```
hey
```

```
hi
```

```
hi
```

```
hi
```

30. different ways to access a linux machine server remotely (from a windows machine)?

=====

using some tools and terminal like

1. putty
2. git bash
3. cmd

***** related to the file permissions *****

31. what are different types of permissions for a file in linux

=====

read (r)
write (w)
executable (x)

32. which permission allow a user to run an executable file (Script)?

=====

we need to provide executable (x) permission to the user

***** Redirection *****

33. how to write the output of a command into a file?

=====

whenever we execute a command we can see the output in a terminal, but the output will redirect into a file

command > file_name

pwd > pwd.txt

```
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$ pwd  
/c/Users/2142289
```

```
CTS+2142289@LTIN531281 MINGW64 ~  
$ pwd > pwd.txt
```

```
CTS+2142289@LTIN531281 MINGW64 ~  
$ ls
```

```
CTS+2142289@LTIN531281 MINGW64 ~  
$ cat pwd.txt
```

/c/Users/2142289

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\$ ls > pwd.txt

CTS+2142289@LTIN531281 MINGW64 ~

\$ ls

34. how to write something in a file without deleting the existing content?

=====

we can append the file using >>

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\$ pwd > pwd.txt

CTS+2142289@LTIN531281 MINGW64 ~

\$ cat pwd.txt

/c/Users/2142289

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\$ pwd >> pwd.txt

CTS+2142289@LTIN531281 MINGW64 ~

\$ pwd >> pwd.txt

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\$ cat pwd.txt

/c/Users/2142289

/c/Users/2142289

/c/Users/2142289

35. how to redirect an error of a command into a file?

=====

if your command is unsuccessful in those cases how to redirect those errors into a file

--> to redirect an error we need to use 2>

to redirect both error and output, 2>&1

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\$ cat abc

cat: abc: No such file or directory

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\$ cat abc 2> abc.txt

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\$ cat abc.txt

cat: abc: No such file or directory

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36. how to automate any task or script?

=====

using cron jobs

for which we have crontab and at command

for the cronjobs we need to use crontab command or at command

37. how to check scheduled jobs?

=====

someone already created the cronjobs or automated skew tasks, how to check those existing jobs

crontab -l

38. what is the meaning of this cron job? * * * * *

=====

this is the pattern which we provide while we creating the cron jobs for automating the tasks

* - minute (0-59)

* - hour (0-23)

* - day of the month (1-31)

* - month (1-12)

* - day of the week (0-6)

39. if your cron job didn't work, how would you check?

=====

for example, if you have scheduled your task and it didn't work? how and where you can check?

check system time,

crontab entry,

check /var/log/messages

***** Services *****

40. what is deamon service?

=====

service that keep running in background
example: httpd, sshd, chronyd

41. how to check if a service is running or not?

=====

lets take an example of httpd service

systemctl status service_name

systemctl status httpd.service

42. how to start/stop any service?

=====

systemctl start service_name

systemctl stop service_name

***** System Monitoring *****

43. how to check for free disk space?

=====

we can use df command

--> df

--> df -h (human readable format)

44. how to check the size of a directory's content?

=====

we can use du command

du - (disk utilization command)

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\$ du new_file

1 new_file

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\$ du -h new_file

1.0K new_file

45. how to check CPU usage for a process?

=====

using top command

top

***** Process *****

46. what is a process in linux?

=====

--> an instance of a running program.

--> whenever you start a program/application or execute a command, a process is created.

--> for every process a unique number is assigned which is called PID (Process ID)

47. how to check if a process/application is running or not?

=====

using ps command

ps - (process status)

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\$ ps

PID	PPID	PGID	WINPID	TTY	UID	STIME	COMMAND
1665	994	1665	27296	pty0	4096	17:14:52	/usr/bin/ps
994	993	994	24260	pty0	4096	02:38:16	/usr/bin/bash
993	1	993	30648	?	4096	02:38:15	/usr/bin/mintty

48. how to terminate/stop a running process?

=====

using kill command

kill PID

49. difference between kill and kill -9?

=====

kill -9 will terminate a program forcefully

***** Networking *****

50. how to check if a IP/Server is accessible or not?

=====

if from your server you are trying to access the other server that IP or server is accessible or not
how you can check

we can use ping or telnet command

ping www.google.com

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\$ ping www.google.com

Pinging www.google.com [172.217.174.68] with 32 bytes of data:

Reply from 172.217.174.68: bytes=32 time=29ms TTL=118

Reply from 172.217.174.68: bytes=32 time=22ms TTL=118

Reply from 172.217.174.68: bytes=32 time=20ms TTL=118

Reply from 172.217.174.68: bytes=32 time=29ms TTL=118

Ping statistics for 172.217.174.68:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 20ms, Maximum = 29ms, Average = 25ms

51. which command to use to get info about ports?

=====

we can use netstat command

netstat -l

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\$ netstat -l

52. how to check open port on linux system?

=====

how to check if a port is available or not available actually are open for connectivity on your
server or not

netstat putan | grep port_no

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\$ netstat putan | grep 22

53. how to check network interfaces in linux?

=====

we can use ifconfig and netstat command

54. difference between Telnet and SSH?

=====

SSH is secured and telnet is not

***** SSH *****

55. which service should be running on server to allow you to connect remotely?

=====

suppose you have a linux server, you want to make your server available to connect remotely

ssh or sshd

systemctl status sshd

56. what is SSH?

=====

SSH or Secure Shell is a network communication protocol that enables two computers/devices to communicate and share data

so in order to enable the communication between 2 servers we need the SSH service.
we have already seen that SSH service is enabled

57. why it is called as Secure Shell?

=====

because communication between host and client will be in encrypted format

58. what is the default port for SSH?

=====

59. which command is used to access a linux system from a terminal or another linux server?

=====

we need ssh command

ssh user@198.169.x.x (IP address)

Archive:

60. if you want to transfer a large file (10GB) remotely, what would be the first thing you will do?

--> to transfer the 10GB file, first thing we can do is to compress the file using gzip command and you can also see the tar and gzip command

--> using these commands we can first compress the file, reduce its size, it can easily transfer to another location or the server and on that server we can decompress or unzip those files using that gunzip command

difference between tar, gzip and gunzip

compress the file using - tar, gzip

unzip the file using - gunzip

1. what is RHEL Linux stands for?

=====

Red Hat Enterprise Linux operating system

it is enterprise, we will get the tech support for any issues and troubleshooting from the company.

2. give some examples o linux distributions?

=====

linux distributions

centOs

ubuntu

fedora

debian

3. what is kernel?

=====

in the linux world kernel is the most important part of linux OS

--> the linux kernel is the main component of a linux OS and is the core interface between a computer's hardware and its processes.

--> it communicates between the 2, managing resources are efficiently as possible.

4. what is swap space?

=====

--> swap space in linux is used when the amount of physical memory (RAM) is full. if the system needs more memory resources and the RAM is full, inactive pages in memory are moved to the swap space.

--> so that we can use the RAM again without any blocker

5. how to search a word in file and replace it in entire file?

=====

using sed command

sed 's/<string_to_change>/<new_string>/g' file_name

s stands for substitue

g stands for globally

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```
$ sed 's/Hi/Hiiii/g' new_file
```

Hiiii, This is sample file

Hiiii

hey

hi

hey

hi

hey

hi

hey

6. what is the use of SCP command?

=====

SCP - secure copy command

--> the SCP command copies files or directories between a local and a remote system or between two remote systems.

--> SCP uses SSH for data transfer.

7. what is the use of FTP command?

=====

FTP is the file transfer protocol to exchange files to and from a remote computer or network

8. what is alias and how to setup?

=====

to create shortcut to the command or task

```
alias d="ls-ltr | awk '{print \$9}'"
```

if the new time you use d the whole big command will execute

9. what is the difference between upgrade and update system?

=====

update:

updates all the presently installed packages to their latest versions that are available in the repositories

upgrade:

performs the same as "yum update", but once finished it also removes all of the obsolete packages from the system.

if you update the package it will update to the latest version of packages and removed the older versions.

10. name default ports used for DNS, SMTP, FTP, SSH, DHCP and squid

=====

DNS - 53
SMTP - 25
FTP - 21
SSH - 22
DHCP - 67, 68
squid - 3128
HTTP - 80
HTTPS - 443

11. how to check if a package is installed or not?

=====

`rpm -qa | grep <package_name>`

12. which of the following commands sends exactly three ping packets to google.com?

=====

`ping -c 3 google.com`

13. which file contains a list of group names and GIDs?

=====

`/etc/group`

`less /etc/group`

14. which of the following contains the value of the exit status of the previously executed command?

=====

`$?`

`echo $?`

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\$ echo \$?

1

15. which command displays the number and type of processors used by a linux system?

=====

cat /proc/cpuinfo

lscpu

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\$ cat /proc/cpuinfo

16. which command should you use to get a list of connected USB devices?

=====

lsusb (answer)

1. listusb

2. usbls

3. lsusb (answer)

4. getusb

17. which of the following environment variables is used as the default argument for the "cd" command when no directory is provided?

=====

Home

1. PWD

2. Home

3. PATH

4. USERPROFILE

18. True or False: A command following a semicolon will be executed no matter the exit status of the command that preceded the semicolon

=====

example:

cd/root; cat/etc/passwd

cd/root; is unsuccessful, the next command will be executed

True

19. what are the valid values of the exit status?

=====

0-255

20. according to the FHS, what is the correct directory for log files?

=====

FHS - file system hierarchy standard

as per the standard where should be the log files presently

/var/log

cd /var/log

21. what port must be open to ping a given host?

=====

ping don't use port

22. how will you transfer file from a linux server to your windows OS and vice versa?

=====

so file transfer between different operating system like from windows to linux, linux to windows

using SSH, FTL, SFTP and tools like WinSCP, FileZilla

23. can you split a file into two? if yes, then how?

=====

split -l 3 file_name

24. how you can get the unique values form the list?

=====

sort file_name | uniq

uniques values will work with the sorted values, first you need to sort the list and then use the unique command to get the unique values

25. using VI editor, how you will edit and save a file?

=====

vi file_name

- - for insert mode

ESC - to come out of insert mode

press :wq - to save the file

26. what is the difference between find and locate command?

=====

locate command search in its own db and you will need to keep updating db

find command search the file in the entire file system

27. what is the TEE command used for?

=====

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\$ pwd | tee new_file

/c/Users/2142289

CTS+2142289@LTIN531281 MINGW64 ~

\$ cat new_file

/c/Users/2142289

Runs command, displays its output on screen, and saves it to file.txt.

28. what is Xargs used for?

=====

Xargs:

it convert the stdInput into command line argument

cat file_name | xargs wc -l

cat new_file | xargs wc -l

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\$ cat new_file | xargs wc -l

wc: /c/Users/2142289: Is a directory

0 /c/Users/2142289

29. how you can find no.of files, folders in a directory?

=====

```
ls -l | wc -l
```

```
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```

```
$ ls -l | wc -l
```

```
42
```

30. I want to read only line 25 to 30th Line, how you will do it?

=====

```
head -30 file_name | tail -5
```

```
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```

```
$ head -30 new_file | tail -5
```

```
Hi, This is sample file
```

```
Hey , How are you
```

```
Hi, have a good day
```

```
Hii, have a nice day
```

```
Hey, sample file
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ head -30 new_file | tail -3
```

```
Hi, have a good day
```

```
Hii, have a nice day
```

```
Hey, sample file
```

31. what is the use of Tar command?

=====

The tar command is used to compress a group of files into an archive. Tar archives combine multiple files and/or directories together into a single file

32. How to redirect both standard output and error to a file?

=====

```
command > file 2>&1
```

```
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```

```
$ pwd > new_file 2>&1
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ cat new_file
```

```
/c/Users/2142289
```

33. name some commands for DNS resolution?

=====

ping, nslookup, dig and host

34. what is the use of At command?

=====

command used to schedule a task once

for example you have created a script and you want to run it tomorrow, we can use the add command

35. which service is used by At command?

=====

Atd (at daemon service)

36. which command syntax is correct if you want to assign read only permission to everyone and write permission to the owner of the file?

=====

chmod u+w, chmod a+r (answer)

chmod o+w, chmod a+r

chmod u+w, chmod a+r (answer)

chmod a+w, chmod o+r

chmod u-w, chmod a+r

37. which command changes the group ownership of the "/var/www" directory to the "nginx" group?

=====

chgrp nginx /var/www (answer)

chgrp nginx /var/www (answer)

changegrp nginx /var/www

grpchange nginx /var/www

38. what is chmod 777 file_name?

=====

This is a way of changing permission using numeric mode.

777 means provide all the permissions rwx to user, group and other users also.

39. what is ACL and its advantage?

=====

Access Control List is used to modify the permissions of files.

for this we use setfacl and getfacl commands

Advantage: we can provide permission to a specific user

40. what are links in linux and its type?

=====

a connection between a file name and the actual data on the disk

we can call it a shortcut

2 types of links in linux

1. hard link

2. soft link

41. difference between hard and soft links?

=====

soft link:

link will be removed if original file removed or deleted

ln -s

hard link:

renaming, deleting or removing the file will not effect the link

ln

42. what command you can use to see all the environment variables?

=====

env or printenv

43: what are the different ways to set environment variables?

=====

at user level:

1. temporary
2. permanent

at global level

44. how can you set environment variables in linux?

=====

using export command

to set variable for the current user - .bashrc

to set variable for globally - /etc/bashrc or /etc/profile

45. what command allows you to display all the process IDs of the running nginx processes?

=====

pidof httpd

46. what is nice value of a process?

=====

nice scale goes from -20 to 19. the lower the no. more priority that task gets.

Process priority = nice

(ex: nice -n 5 process)

47. how to check the nice value of a process?

=====

ps -l PID

48. you want a process with a pID of 8675 to complete its task faster than the default rate. which command should you run to accomplish this goal?

=====

renice -20 -p 8675

this can be done by top command also

49. what command can be used to check the %CPU and %Memory of a process?

=====

top command

50. if you want to run a task/script in background even if you close your terminal, what is the way?

=====

using nohup command

nohup process &

nohup process > /dev/null 2>&1 &

51. how to see all the active jobs?

=====

jobs

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\$ jobs

CTS+2142289@LTIN531281 MINGW64 ~

\$ sleep 5s

[1]+ Stopped sleep 5s

CTS+2142289@LTIN531281 MINGW64 ~

\$ jobs

[1]+ Stopped sleep 5s

52. how to resume a stopped job in background or foreground?

=====

fg

jobs

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\$ fg

sleep 5s

CTS+2142289@LTIN531281 MINGW64 ~

\$ jobs

53. which command displays information about kernel-related messages along with hardware and system startup messages stored in kernel ring buffer?

=====

somehow useful especially for the admin people to troubleshoot or do analysis

dmesg

54. in a directory you have a combination of files like xml, yml, scripts, logs etc, how you will get/see only xml files?

=====

ls -l *.xml

ls -l *.txt

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\$ ls -l *.txt

-rw-r--r-- 1 CTS+2142289 4096 36 Nov 21 16:40 abc.txt

-rw-r--r-- 1 CTS+2142289 4096 23 Nov 21 12:25 file1.txt

-rw-r--r-- 1 CTS+2142289 4096 51 Nov 21 16:34 pwd.txt

55. create 100 files with naming file1, file2, file3, ... file 100

=====

touch file{1...100}

CTS+2142289@LTIN531281 MINGW64 ~/file

\$ touch file{1...100}

56. how would you display all the files in the present working directory that start with either the letter "c" or "m"?

=====

ls [cm]*

ls [fn]*

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\$ ls [fn]*

file1.txt new_file newfile3 ntuser.dat.LOG1 ntuser.dat.LOG2 ntuser.ini

file:

file{1...100}

57. run a command that shows all the lines except any lines starting with the a character # in a file?

=====

```
cat file | grep -v ^#
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ cat new_file
```

```
Hi, This is sample file
```

```
Hey , How are you
```

```
Hi, have a good day
```

```
Hii, have a nice day
```

```
Hey, sample file
```

```
# Hi, this is comment
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ cat new_file | grep -v ^#
```

```
Hi, This is sample file
```

```
Hey , How are you
```

```
Hi, have a good day
```

```
Hii, have a nice day
```

```
Hey, sample file
```

58. which of the following commands lists all files and directories except those beginning with the letter "a"?

=====

```
ls -l | grep -v ^a (answer)
```

```
1. ls -l | grep -i ^a
```

```
2. ls -l | grep -v ^a (answer)
```

```
3. ls -l | grep -e ^a
```

```
4. ls -l | grep -x ^a
```

59. which of the following bash commands produces the output of "bat bet bit bot"?

=====

```
echo b{a,e,i,o}t (answer)
```

```
1. echo b{ae}t
```

```
2. echo b{a,e,i,o}t (answer)
```

```
3. echo b[a,e,i,o]t
```

```
4. echo b[aeio]t
```

```
CTS+2142289@LTIN531281 MINGW64 ~
```

```
$ echo b{a,e,i,o}t
```

bat bet bit bot

60. what is a VM?

=====

A virtual machine (VM) is a virtual environment that functions as a virtual computer system with its own CPU, memory, network interface, and storage, created on a physical hardware system

61. you have a windows machine and you want to test an application on Linux, what you can do?

=====

we can use VM to use multiple OS like Linux on our windows machine and test our application

62. what is hypervisor?

=====

it is software that creates and runs virtual machines (VMs)

63. what are the type of hypervisor?

=====

types of hypervisor are

hosted

bare metal hypervisor

64. which command can you use to display a list of currently configured logical volumes?

=====

lvs

or sudo lvs (as it required root access)

1. lv

2. lvs (answer)

3. vls

4. vl

65. which command can you use to display a list of currently configures volume groups?

=====

vgs (answer)

1. vg
2. vgs (answer)
3. lv
4. vgl

66. which command displays memory usage, including the amount of swap space being used?

=====

free
free -h

67. the /home partition is running out of disk space. which command can you use to determine which user's home directory is using the most space?

=====

we can use du command

68. what is linux filesystem?

=====

OS store and manage data on disk or partitions using a structure called filesystem.

filesystem includes files, directories and its related permissions.

69. how to check your linux filesystem?

=====

lsblk -f

70. how do you create new user in Linux?

=====

you can create a new user in linux using the useradd or adduser command.

```
sudo useradd -m -d /home/newuser -s /bin/bash -c "New User" -u 1001 -g users -G sudo,adm -e 2025-01-01 new user
```

71. what is the root user in linux?

=====

the root user is the superuser in linux, with full administrative privileges to the entire system.

root user
/root home dir

/root dir

72. explain the file permissions in linux?

=====

linux file permissions include read(r), write(w), and execute(x) for three categories: owner, group, and others

ls -l file_name

```
[root@ip-172-31-24-117 /]# ls -l prasanna.txt
-rw-r--r--. 1 root root 73 Nov 22 09:53 prasanna.txt
```

73. how do you change file permissions in linux?

=====

file permissions are changes using the chmod command

chmod u=rwx,g=rx,o= file.txt

chmod u=rwx,g=rw,o=rw prasanna.txt

```
[root@ip-172-31-24-117 /]# chmod u=rwx,g=rw,o=rw prasanna.txt
[root@ip-172-31-24-117 /]# ls -l prasanna.txt
-rwxrw-rw-. 1 root root 73 Nov 22 09:53 prasanna.txt
```

74. how do you view active processes in linux?

=====

you can view active processes using commands like ps, top, or htop

ps
top
htop

75. how do you kill a process in linux?

=====

A process can be killed using the kill command followed by the process ID

kill PID

76. what is the use of the df command in linux?

=====

the df command is used to display disk space usage of file systems

df -h

```
CTS+2142289@LTIN531281 MINGW64 ~/OneDrive - Cognizant/Documents/AWS
```

```
$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
C:/Users/2142289/AppData/Local/Programs/Git	475G	182G	293G	39%	/

```
CTS+2142289@LTIN531281 MINGW64 ~/OneDrive - Cognizant/Documents/AWS
```

```
$ df
```

Filesystem	1K-blocks	Used	Available	Use%	Mounted on
C:/Users/2142289/AppData/Local/Programs/Git	497602556	190375232	307227324	39%	/

77. what does the free command do in linux?

=====

the free command displays the total amount of free and used physical RAM and swap memory in the system

```
[ec2-user@ip-172-31-24-117 ~]$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	949Mi	127Mi	598Mi	0.0Ki	223Mi	684Mi
Swap:	0B	0B	0B			

```
[ec2-user@ip-172-31-24-117 ~]$ free
```

	total	used	free	shared	buff/cache	available
Mem:	972156	130856	612504	468	228796	700468
Swap:	0	0	0			

78. explain the purpose of the grep command

=====

grep is used to search text or searches the given file for lines containing a match to the given strings or words

```
grep "string file.txt"
```

```
[ec2-user@ip-172-31-24-117 /]$ grep "nice" prasanna.txt
```

```
Hey, have a nice day
```

79. what is the purpose of the crontab in linux?

=====

crontab is used to schedule commands to be executed periodically. it holds the list of scheduled tasks

```
[ec2-user@ip-172-31-24-117 /]$ crontab -l
-bash: crontab: command not found
```

```
[ec2-user@ip-172-31-24-117 /]$ sudo yum install cronie
```

```
[ec2-user@ip-172-31-24-117 /]$ crontab -l
no crontab for ec2-user
```

80. how can you find the IP address of a linux server?

=====

you can find the IP address using commands like ifconfig or ip addr show

ifconfig
ip addr

```
[ec2-user@ip-172-31-24-117 /]$ ifconfig
enX0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 172.31.24.117 netmask 255.255.240.0 broadcast 172.31.31.255
    inet6 fe80::8ff:feff:fe71:3c1b prefixlen 64 scopeid 0x20<link>
    ether 0a:ff:fe:71:3c:1b txqueuelen 1000 (Ethernet)
    RX packets 8741 bytes 2100717 (2.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 7155 bytes 770355 (752.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 12 bytes 1020 (1020.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 12 bytes 1020 (1020.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
[ec2-user@ip-172-31-24-117 /]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group
default qlen 1000
```

```
link/ether 0a:ff:fe:71:3c:1b brd ff:ff:ff:ff:ff:ff
altname eni-0d4144f8b8352e459
altname device-number-0.0
inet 172.31.24.117/20 metric 512 brd 172.31.31.255 scope global dynamic enX0
    valid_lft 3033sec preferred_lft 3033sec
inet6 fe80::8ff:feff:fe71:3c1b/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
```

81 explain the purpose of the sshd service?

=====

sshd is the SSH daemon that allows secure remote connections to the server

82. how do you check the status of a service in linux?

=====

the status of a service can be checked using

systemctl status servicename

```
[ec2-user@ip-172-31-24-117 /]$ systemctl status sshd
```

```
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-22 09:42:50 UTC; 2h 13min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 2117 (sshd)
    Tasks: 1 (limit: 1106)
   Memory: 9.2M
      CPU: 685ms
  CGroup: /system.slice/ssh.service
          └─2117 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

83. what is the purpose of the etc/passwd file?

=====

the etc/passwd file contains user account information

```
[ec2-user@ip-172-31-24-117 /]$ cat /etc/passwd
```

```
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
```

halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
prasanna:x:1001:1001::/home/prasanna:/bin/bash
lakshmi:x:1002:1002::/home/lakshmi:/bin/bash

84. how do you search for a file in linux?

=====

you can search for a file using commands like find or locate

```
[ec2-user@ip-172-31-24-117 /]$ find prasanna.txt  
prasanna.txt
```

85. what are environment variables in linux?

=====

an environment variable is a name value that can affect the way running processes behave on a computer

they are part of the environment in which a process runs

printenv

86. how do you set an environment variable in linux?

=====

environment variables can be set using the export command,

eg: export VARNAME=value

87. what is the purpose of the /etc/shadow file?

=====

the /etc/shadow file stores actual password data in an encrypted format for user's accounts with additional properties related to user password

```
[ec2-user@ip-172-31-24-117 /]$ cat /etc/shadow
cat: /etc/shadow: Permission denied
```

```
[ec2-user@ip-172-31-24-117 /]$ sudo cat /etc/shadow
```

88. explain the use of the ping command?

=====

ping is used to test the reachability of a host on an IP network and to measure the round-trip time for messages sent to the destination computer

ping www.google.com

```
[ec2-user@ip-172-31-24-117 /]$ ping www.google.com
PING www.google.com (142.251.111.104) 56(84) bytes of data.
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=1 ttl=106 time=1.62 ms
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=2 ttl=106 time=2.01 ms
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=3 ttl=106 time=1.87 ms
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=4 ttl=106 time=1.62 ms
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=5 ttl=106 time=1.61 ms
64 bytes from bk-in-f104.1e100.net (142.251.111.104): icmp_seq=6 ttl=106 time=1.63 ms
^C
```

--- www.google.com ping statistics ---

6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 1.606/1.724/2.007/0.156 ms

89. how do you check disk usage in linux?

=====

disk usage can be checked using the df and du commands

df, df -h
du -h filename

```
[ec2-user@ip-172-31-24-117 /]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           475M   0  475M   0% /dev/shm
```

```
tmpfs      190M 460K 190M  1% /run
/dev/xvda1  8.0G 1.7G 6.4G 21% /
tmpfs      475M  0 475M  0% /tmp
/dev/xvda128 10M 1.3M 8.7M 13% /boot/efi
tmpfs      95M  0 95M  0% /run/user/1000
```

```
[ec2-user@ip-172-31-24-117 ~]$ du -h prasanna.txt
4.0K  prasanna.txt
```

90. explain the use of the tail command

=====

tail displays the last part of files. it's commonly used to view the end of log files

tail -5 filename

91. what is the purpose of the xargs command?

=====

xargs is used to build and execute command lines from standard input. its useful for combining commands in a pipeline

find . -name "*.txt" | xargs rm

92. how do you schedule a task in linux?

=====

tasks can be scheduled using cron jobs, which are defined in crontab files

*/15 * * * * /path/to/command

93. how do you list all installed packages in linux?

=====

this can be done using the package manager's list command, like

apt list --installed in Debian/Ubuntu or
rpm -qa in RHEL/CentOS

```
[ec2-user@ip-172-31-24-117 ~]$ rpm -qa
```

94. how do you start, stop and restart services in linux?

=====

services are managed using the systemctl command

eg:

systemctl start servicename

systemctl stop servicename and

systemctl restart servicename

95. explain the purpose of the /etc/fstab file

=====

/etc/fstab is a system configuration file that contains information about different file systems and how they are mounted

sudo cat /etc/fstab

```
[ec2-user@ip-172-31-24-117 ~]$ sudo cat /etc/fstab
```

```
#
```

```
UUID=9a57ac2f-bfe7-4e29-bf89-56caddc22c97 / xfs defaults,noatime 1 1
```

```
UUID=9682-52BE /boot/efi vfat
```

```
defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2
```

96. how do you create a simple script in linux?

=====

a simple script can be created using a text editor, starting with the sheband line (#!/bin/bash for a bash script), followed by shell commands. the script file should be made executable with `chmod +x scriptname`.

vi test.sh

in vi editor

```
#!/bin/bash
```

```
echo HELLO
```

```
:wq
```

two ways to execute this command

```
bash test.sh
```

```
chmod +x test.sh
```

```
./test.sh
```

```
[ec2-user@ip-172-31-24-117 ~]$ vi test.sh
```

```
#!/bin/bash
```

```
echo HELLO
```

```
[ec2-user@ip-172-31-24-117 ~]$ bash test.sh
```

```
HELLO
```

```
[ec2-user@ip-172-31-24-117 ~]$ chmod +x test.sh
```

```
[ec2-user@ip-172-31-24-117 ~]$ ./test.sh
HELLO
```

97. what is the use of the nohup command?

=====

nohup is used to run a command/script in the background even after logging out from the shell

98. what is the use of the sed command?

=====

sed (stream editor) is used for parsing and transforming text, typically used in pipelines

sed 's.Linux/UNIX/g' example.txt

```
[ec2-user@ip-172-31-24-117 /]$ cat prasanna.txt
Hi, this is sample text file
Hey, have a nice day
hello, have a good day
[ec2-user@ip-172-31-24-117 /]$ sed 's/nice/wonderful/g' prasanna.txt
Hi, this is sample text file
Hey, have a wonderful day
hello, have a good day
```

99. what is the purpose of the /var directory?

=====

the /var directory contains variable data like system logs (/var/log), mail(/var/mail), and temporary files required for running applications

```
[ec2-user@ip-172-31-24-117 /]$ cd var
[ec2-user@ip-172-31-24-117 var]$ cd log
[ec2-user@ip-172-31-24-117 log]$ ls
README  btmp          cloud-init.log  dnf.rpm.log    lastlog  sssd
amazon  chrony        dnf.librepo.log hawkey.log     private  tallylog
audit   cloud-init-output.log dnf.log        journal        sa       wtmp
```

```
[ec2-user@ip-172-31-24-117 log]$ cd ..
[ec2-user@ip-172-31-24-117 var]$ cd mail
[ec2-user@ip-172-31-24-117 mail]$ ls
ec2-user lakshmi prasanna rpc
```

100. how do you monitor system performance in linux?

=====

system performance can be monitored using tools like

top
htop
vmstat
iostat, and sar

101. explain the purpose of the ssh command?

=====

ssh (Secure Shell) is a protocol used to securely access and manage systems remotely

102. explain the use of the awk command?

=====

awk is a powerful programming language and tool used for pattern scanning and processing

awk '{print \$2}' data.txt

```
[ec2-user@ip-172-31-24-117 /]$ cat prasanna.txt
Hi, this is sample text file
Hey, have a nice day
hello, have a good day
```

```
[ec2-user@ip-172-31-24-117 /]$ awk '{print $2}' prasanna.txt
this
have
have
```

103. how do you find the amount of free memory in the system?

=====

free memory can be checked using the free command or by examining the contents of /proc/meminfo

104. how do you create a swap file in linux?

=====

a swap file can be created using
dd to allocate space,
mkswap to set up a linux swap area, and
swapon to enable the swap file

105. What is dmidecode command used for?

=====

The dmidecode command in Linux is used to retrieve detailed information about the system's hardware from the BIOS/DMI table in a human-readable format.

```
sudo yum install dmidecode
```

```
[ec2-user@ip-172-31-24-117 /]$ sudo dmidecode | more
# dmidecode 3.6
Getting SMBIOS data from sysfs.
SMBIOS 2.7 present.
11 structures occupying 378 bytes.
Table at 0x000EB01F.
```

106. What are runlevels in Linux?

=====

Runlevels are predefined modes that a Linux system can run in, each with a specific purpose, like single-user mode, multiuser mode, GUI mode, etc. check using runlevel command

0: Halt/Shut down the system

5: Full multi-user mode (with graphical user interface)

```
[ec2-user@ip-172-31-24-117 /]$ runlevel
N 5
```

107. How do you change runlevels?

=====

Runlevels can be changed using the init/telinit command or systemctl in systemd-based systems.

108. How do you change the hostname of a Linux system?

=====

The hostname can be changed using the hostnamectl command or by editing /etc/hostname.

```
vim /etc/hostname
```

109. What is the purpose of the /etc/cron.* directories?

=====

These directories contain scripts that are run periodically at fixed times, dates, or intervals by the cron daemon.

110. How do you find out which processes are listening on a specific port?

=====

This can be found using netstat -tuln or ss -tuln.

111. How do you check the kernel version of your Linux system?

=====

The kernel version can be checked using uname -r.

```
[ec2-user@ip-172-31-24-117 /]$ uname -r  
6.1.158-178.288.amzn2023.x86_64
```

112. : How do you create a read-only file in Linux?

=====

A file can be made read-only using the chmod command, for example, chmod 444 filename.

113. What is Linux and what are its main features?

=====

Linux is a free and open-source operating system based on Unix. Its main features include its stability, security, flexibility, free of cost, and support for a wide range of hardware.

114. What is the difference between Linux and UNIX?

=====

The primary difference is that Linux is open source and free, while UNIX is proprietary. Additionally, Linux is widely used for a variety of applications, whereas UNIX is primarily used in large enterprises.

115. What is the Linux Kernel?

=====

The Linux kernel is the core of the Linux operating system. It acts as an interface between the computer's hardware and the

software running on it.

It manages hardware resources, enables process control, file management, and system security.

116. What is a shell in Linux?

=====

A shell in Linux is an interface that allows users to interact with the operating system by running commands.

117. What are some common shells in Linux?

=====

Common Linux shells include Bash (Bourne-Again Shell), Tcsh, Zsh, and Ksh (Korn Shell).

118. What is a process in Linux?

=====

A process is an running instance of a program in Linux. Each process in Linux has a unique identifier called a Process ID (PID).

119. What is a daemon in Linux?

=====

A daemon is a background process that runs continuously and typically starts at boot.

120. What is the difference between soft link and hard link in Linux?

=====

A soft link is a pointer to a file, while a hard link is a direct mirror of the file. Deleting the original file affects a soft link but not a hard link.

121. How do you compress and extract files in Linux?

=====

Files can be compressed using tools like gzip, bzip2, and extracted using gunzip, bunzip2.

122. What is a symbolic link in Linux?

=====

A symbolic link, or symlink, is a file that points to another file or directory. It's a reference that can be used to access the target file.

123. What are inodes in Linux?

=====

An inode is a data structure on a filesystem that stores information about a file except its name or actual data.

Every file or directory is associated with an inode, which is identified by an inode number in the filesystem. This inode number is a unique identifier within the filesystem.

124. Explain the use of the vi editor.

=====

vi is a text editor in Linux, used for creating and editing files.

125. What is the purpose of the iptables?

=====

iptables is a command-line utility in Linux used for configuring network packet filtering rules.

With iptables, you can define rules for how incoming, outgoing, and forwarding network packets should be handled and manipulated.

126. What is the difference between wget and curl?

=====

Both are command-line tools for transferring data over the network.

wget is simpler and designed to download files, whereas curl offers more protocols and features.

protocols including HTTP, HTTPS, FTP, FTPS, SCP, SFTP, TFTP, LDAP, LDAPS, DICT, TELNET, FILE, IMAP, POP3, SMTP, and SMB.

127. How do you schedule a task in Linux?

=====

Tasks can be scheduled using cron jobs, which are defined in crontab files.

`*/15 * * * * /path/to/command`

128. What is a package manager in Linux?

=====

A package manager is a collection of tools that automates the process of installing, upgrading, configuring, and removing software packages in a consistent manner.

129. Name some common package managers in Linux.

=====

Common package managers include apt (for Debian/Ubuntu), yum (for RHEL/CentOS), and zypper (for openSUSE).

130. What is the difference between yum and rpm?

=====

yum is a front-end tool for package management that automatically resolves dependencies, while rpm is a lower-level tool that manages individual packages without automatically handling dependencies.

140. What is a Linux service?

=====

A service in Linux is a program that runs in the background, typically initiated at system startup.

141. What is LVM and what are its advantages?

=====

LVM stands for Logical Volume Manager. It allows for flexible disk management by creating, resizing, and deleting logical volumes.

142. : How do you secure an SSH server?

=====

SSH security can be enhanced by disabling root login, changing the default port, using key-based authentication, and configuring firewalls.

143. What is the purpose of the /home directory?

=====

The /home directory contains the personal directories of all users.

144. What is swap space in Linux?

=====

Swap space is a portion of a hard disk drive (HDD) that is used for virtual memory when the physical memory (RAM) is full.

145. What is the purpose of the /etc/hosts file?

=====

The /etc/hosts file is used to map hostnames to IP addresses.

146. How do you set a static IP address in Linux?

=====

A static IP address is set by editing network configuration files or using network management tools or commands, depending on the distribution.

Ex: nmcli

147. Explain the purpose of the nslookup command?

=====

nslookup is a network administration command-line tool used for querying the Domain Name System (DNS) to obtain domain name or IP address mapping.

148. What is the difference between iptables and firewalld?

=====

iptables is a traditional form of Linux firewall, and firewalld is a more dynamic, front-end friendly tool that interfaces with iptables configurations.

149. What is the purpose of the /boot directory?

=====

The /boot directory contains essential files needed to boot

the system, including the Linux kernel, initial RAM disk images, and the bootloader configuration.

150. Explain the purpose of the rsync command?

=====

rsync is used for fast and versatile file copying, both locally and remotely. It's commonly used for backups and mirroring.

151. What is a Linux distribution?

=====

A Linux distribution is a version of the operating system that includes the Linux kernel, a package management system and other supporting software and libraries.

152. Name some popular Linux distributions?

=====

Popular distributions include Ubuntu, Fedora, CentOS, Debian, Red Hat Enterprise Linux (RHEL), and openSUSE.

153. What are system logs and where are they located?

=====

System logs record various activities of the operating system. They are typically located in the /var/log directory.

154. Explain the purpose of the netstat command?

=====

netstat is used to display network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.

155. How do you add a new disk to a Linux system without rebooting?

=====

Scan the SCSI bus using commands like
echo "- - -" > /sys/class/scsi_host/host0/scan
Partitioning (ex /dev/sdb using fdisk)
Formatting the new disk (using mkfs)
Mount.

156. What is the purpose of the /etc/sysconfig directory?

=====

/etc/sysconfig contains scripts and configuration files that manage the startup of services and configure aspects of system behavior.

157. How do you update all packages in Linux?

=====

This can be done using the package manager, e.g., apt-get update && apt-get upgrade for Debian/Ubuntu or yum update for RHEL/CentOS.

158. What are the common network configuration files in Linux?

=====

Common files include /etc/network/interfaces (Debian/Ubuntu), /etc/sysconfig/network-scripts/ifcfg-eth0 (RHEL/CentOS), and /etc/netplan/*.yaml (newer Ubuntu versions).

159. : How do you change the default runlevel?

=====

The default runlevel can be changed by editing the /etc/inittab file in SysVinit systems or by using systemctl set-default in systemd systems.

Runlevel 0 (Halt): Mapped to poweroff.target

Runlevel 1 (Single User Mode): Mapped to rescue.target

Runlevel 2 (Multi-user Mode without Networking): Mapped to multi-user.target (but in most modern distributions, networking is generally started at all multi-user runlevels)

Runlevel 3 (Multi-user Mode with Networking): Mapped to multiuser.target

Runlevel 4 (User-Defined/Spare): Mapped to multi-user.target as well, typically not used and available for custom configurations

Runlevel 5 (Multi-user Mode with Networking and GUI): Mapped to graphical.target

Runlevel 6 (Reboot): Mapped to reboot.target

160. What is the purpose of the /etc/resolv.conf file?

=====

/etc/resolv.conf contains information about DNS servers and search domains for the host.

161. How do you find which process is using a particular file?

=====

This can be found using the lsof command, which lists open files and the processes using them.

162. : What is the purpose of the dmesg command?

=====

dmesg displays the kernel-related messages and helps in troubleshooting hardware and driver related issues.

163. Explain the purpose of the usermod command?

=====

usermod is used to modify a user's system properties, such as username, home directory, and group.

164. How do you find the size of a directory in Linux?

=====

The size of a directory can be found using the du command.

165. How do you change the priority of a process in Linux?

=====

The priority of a process can be changed using the renice command.

166. What is the purpose of the mount command?

=====

mount is used to attach a filesystem to a directory in the filesystem tree.

167. Explain the purpose of the umask command?

=====

umask sets the default file creation permissions for new files and directories.

168. What is the purpose of the /opt directory?

=====

/opt is used for storing optional software and packages that are not part of the default installation.

169. Explain the purpose of the dig command?

=====

dig (Domain Information Groper) is a network administration command-line tool for querying DNS name servers.

170. What is quota in Linux?

=====

A quota is a system feature used to limit the amount of disk space and/or the number of files (inodes) a user or group can use.

171. How do you set a quota for users in Linux?

=====

Quotas are set using the quota and edquota commands, and by modifying filesystem configuration.

172. What is the purpose of the /usr directory?

=====

/usr contains user utilities and applications, supporting files for the operating system.

173. How do you find the hardware details of your system?

=====

Hardware details can be found using commands like lshw, lscpu, lsblk, and lspci.

174. What is the purpose of the sysctl command?

=====

sysctl is used to modify kernel parameters at runtime

175. What is the difference between a service and a daemon in Linux?

=====

A daemon is a background process that runs continuously, often started at boot time. A service is a process that performs a specific function and may not run continuously.

176. How do you redirect output in Linux?

=====

Output can be redirected using the > for standard output and 2> for standard error.

177. What is the purpose of the /proc directory?

=====

/proc is a virtual filesystem that provides a mechanism to access kernel and process information in a file-like structure.

178. Explain the purpose of the tr command?

=====

tr is used for translating or deleting characters from standard input and writing to standard output.
echo "hello world" | tr 'a-z' 'A-Z'

179. What is the difference between PATH and LD_LIBRARY_PATH?

=====

PATH is an environment variable specifying a set of directories where executable programs are located.
LD_LIBRARY_PATH is used to specify a set of directories where libraries should be searched for first, before the standard set of directories.